

Practice Quiz: The Inventive Pitch

Part A: Multiple Choice

1. Invention communication in Active Inference is best understood as: A) Transmitting data from sender to receiver without loss B) Acting to update the audience's generative model to align with the inventor's C) Repeating the message until the audience memorizes it D) Providing all available information and letting the audience sort it out
2. The "curse of knowledge" in inventor communication occurs because: A) Inventors deliberately withhold information from audiences B) The inventor's deeply internalized model makes it hard to predict what the audience lacks C) Audiences are not intelligent enough to understand inventions D) Communication technology is insufficient for complex ideas
3. The six-part invention narrative manages the audience's understanding by: A) Presenting all information simultaneously for maximum impact B) Sequencing information so each piece is comprehensible given the current model state C) Avoiding technical details entirely D) Using emotional manipulation rather than evidence
4. Visual communication (diagrams, storyboards) is effective because: A) People prefer pictures to words in all situations B) Visuals serve as externalized model artifacts that the audience's perceptual system can process directly C) Patent law requires visual documentation D) Visuals are cheaper to produce than written documents
5. Patent independent claims describe: A) Every possible variation of the invention B) The commercial potential and market size C) The Markov blanket of the invention — the minimal essential structure D) The inventor's personal background and qualifications
6. Adapting communication for different audiences requires: A) Using the same message for all audiences to maintain consistency B) Modeling the audience's prior beliefs and precision weightings, then designing messages that bridge the gap C) Always leading with the most technical details first D) Focusing exclusively on financial returns regardless of audience type

7. Airbnb's initial pitches failed because: A) The product was not yet built B) Investors' prior models predicted strangers would not share living spaces — the founders presented the full concept at once rather than managing prediction errors sequentially C) The founders lacked presentation skills D) The market was too small for investor interest

Part B: Short Answer / Analysis

1. An engineer has invented a novel water purification system using graphene membranes. She needs to pitch it to a group of venture capital investors who have backgrounds in finance, not engineering. The investors' generative model includes strong beliefs that water purification is a commoditized market with low margins. Design a pitch strategy that accounts for the investors' prior model. Specify how you would sequence information to manage prediction errors, and identify two specific model updates you need to achieve.
2. Steve Jobs introduced the iPhone by first establishing that existing smartphones had problems (they were not smart enough, they were hard to use, they could not do multiple things well). Analyze this as the "tension" phase of the invention narrative using Active Inference concepts. Why was it important to establish the prediction error (current phones fail) before presenting the solution (iPhone)? What would have happened if Jobs had simply shown the iPhone without this setup?
3. You need to communicate your invention to three different audiences in one day: a potential co-inventor (technical peer), a grant committee (evaluators of societal impact), and a group of potential end users. For each audience, describe their prior model of the problem, identify the single most important model update you need to achieve, and specify which communication format (narrative, demo, diagram, data, analogy) would be most effective and why.